The Problem & Our Solution

The problem

- Multi-depot fleets, hundreds of orders
- Deadlines, limited stock and capacity
- Result: extra km, late and unserved orders

Our solution

- Assigns orders to depots and vehicles
- Builds routes, replans live on events
- One HTML file: no install, no backend

5
data streams

5
stress tests

7
invariant checks

1
HTML file

How the Engine Works

Sort	Assign	Insert	2-opt	Score
by deadline	stock + capacity	cheapest position	up to 5 passes	cost + metrics

Optimized algorithm

- Deadline-first cheapest insertion
- Stability penalty limits reshuffling
- Complexity ≈ $O(n \cdot V \cdot L^2)$ insertion

Baseline & scoring

- Nearest depot + nearest-neighbour
- Same cost: km + late + unserved
- Both heuristic, not provably optimal

Live Replanning & Stress Tests

Scenario	What changes	Expected outcome
NORMAL	30 orders, full fleet	Routes, time windows, cost
ORDER RUSH	20 extra orders	Overflow flagged with reasons
BREAKDOWN	Vehicle with stops fails	Orders moved, before/after routes
SHORTAGE	Depot stock cut	Reassigned or flagged
MULTIPLE	Breakdown + rush + shortage + traffic	One replan, event timeline

Every event: record → update data → keep completed → release unfinished → re-solve → explain

Dashboard & Innovation

Live map

Colored routes, arrows, unserved and broken markers

Vehicle animation

Vans and trucks drive routes; deliveries complete live

Tap to track

Line from your touch to the vehicle with load and ETA

Explainable plans

Why each order got its vehicle, plus bottlenecks

Dashboard panels

KPIs, inventory, vehicles, filters, event timeline

Honest comparison

Baseline vs optimized; negative gains shown

Status, Limits & Demo Plan

Implemented in the app

- 5 data streams + live events
- Assignment, routing, 2-opt, replanning
- 5 stress tests + baseline compare
- 7 invariant checks, 10 self-tests

Known limits

- Travel times are simulated
- Heuristics, not provably optimal
- Penalties editable in code only
- Vehicles stop at last customer

Demo: NORMAL → RUSH → BREAKDOWN (tap the vehicle) → SHORTAGE → MULTIPLE → live order → run tests